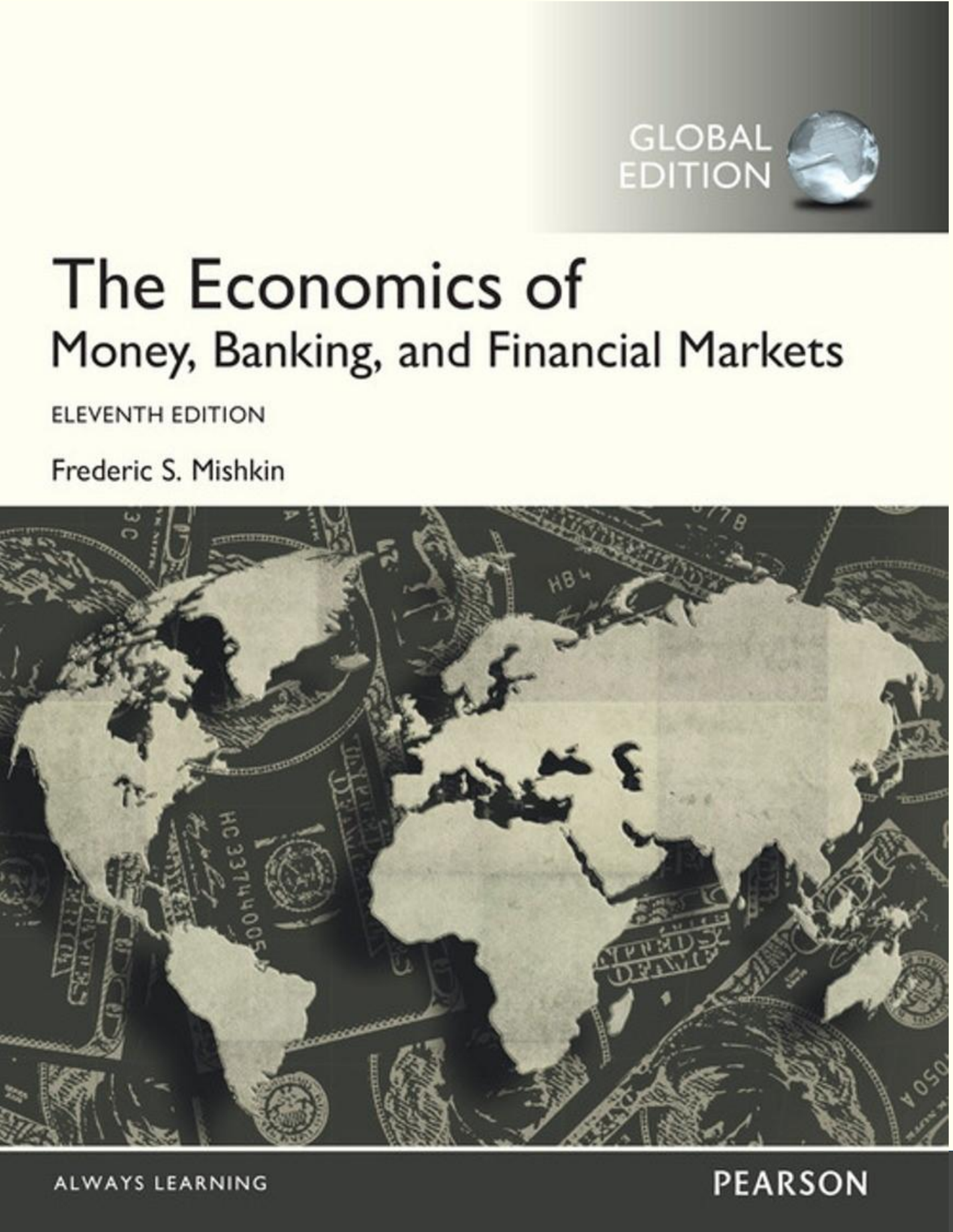
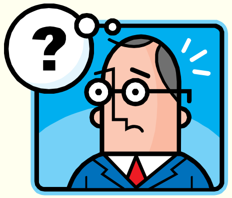


Chapter 3
What Is Money?

Learning Objectives

- Describe what money is
- List and summarize the functions of money
- Identify different types of payment systems
- Compare and contrast the M1 and M2 money supplies

Meaning of Money



What is money?

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Meaning of Money

- Money (a stock concept) is different from:
  - **Currency:** paper money + coins
  - **Wealth**: the total collection of pieces of property that serve to store value
  - **Income**: flow of earnings per unit of time (a flow concept)

Meaning of Money

- The Definition of ***Money*** (or the "money supply"): in Economics

  anything that is generally accepted in payment for goods or services or in the repayment of debts

Functions of Money

- Medium of Exchange
- Unit of Account
- Store of Value

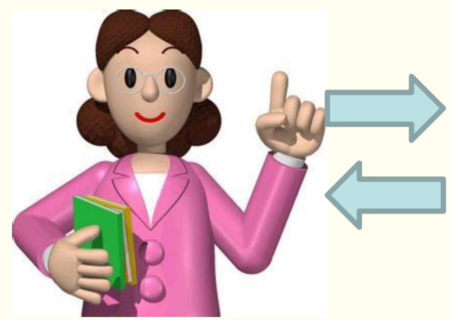
Functions of Money

- **Medium of Exchange**

barter economy:

Barter is the exchange of personal possessions of value for other goods that you want.

"double coincidence of wants"



- **Money**
  - **reducing transaction costs**
  - **encouraging specialization**

Transaction cost!

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Functions of Money

- **Medium of Exchange**

Criteria :

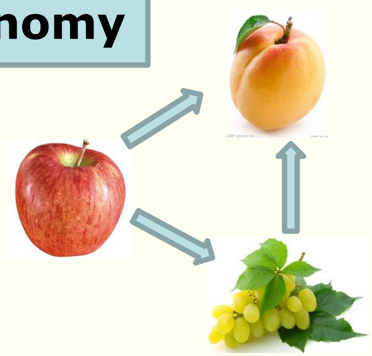
1. easily standardized-- simple to ascertain its value
2. widely accepted
3. divisible-- easy to “make change”
4. easy to carry
5. must not deteriorate quickly

Functions of Money

- **Unit of Account**

Money is used to measure value in the economy

Barter Economy



3 goods: 3 prices

10 goods: 45 prices

100 goods: 4,950 prices

1000 goods: 499,500 prices

- Money reduces transaction costs in an economy by reducing the number of prices that need to be considered

Functions of Money

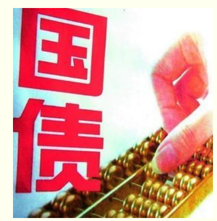
- **Store of Value**

Money:

a repository of purchasing power over time

Liquidity !

the relative ease and speed with which an asset can be converted into a medium of exchange



Money is the most liquid of all assets but loses value during inflation.

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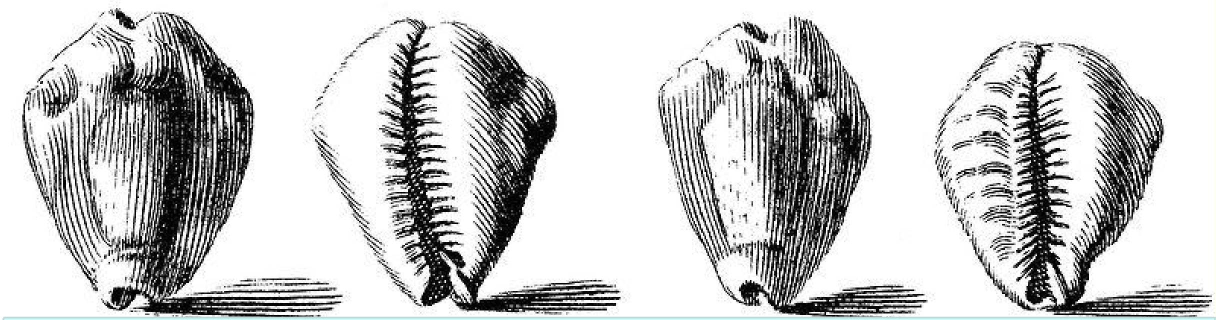
Evolution of the Payments System

- **Commodity Money**

  Money made up of precious metals or another valuable commodity is called commodity money.



1200 B.C. in China

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Evolution of the Payments System

- **Commodity Money**

Raw metals

Problems with commodity money:

very heavy and is hard to transport from one place to another

Evolution of the Payments System

- **Standardized coinage**

Egypt, 1580–1200 B.C.

Colombia, 15th century

Evolution of the Payments System

- **Paper currency**

北宋交子

大明宝钞

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Evolution of the Payments System

- **Paper currency**

  1.Representative money（代用货币）

The term ‘representative money’ has been used in the past “to signify that a certain amount of bullion was stored in a Treasury while the equivalent paper in circulation”

Evolution of the Payments System

- **Paper currency**

  2. Fiat money (不兑现货币、流通券)

  paper currency decreed by governments as legal tender (meaning that legally it must be accepted as payment for debts) but not convertible into coins or precious metal

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Evolution of the Payments System

- **Fiat Money**
  - **legal tender**

Advantages:

much lighter than coins or precious metal

Drawbacks:

easily stolen and can be expensive to transport in large amounts because of their bulk

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Evolution of the Payments System

- **Checks**

  A ***check*** is an instruction from you to your bank to transfer money from your account to someone else's account when she deposits the check.

- n

- ther.

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Evolution of the Payments System

- **Electronic Payment**

Pay your bills electronically !

Evolution of the Payments System

- **E-Money**

  Electronic money (or e-money): money that exists only in electronic form

1. Debit card
2. Stored-value card
3. E-cash

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Cryptocurrency

CBDC

EuroChain

Jasper

Sand Dollar

Ubin

Stella

Evolution of the Payments System

Commodity Money

Fiat Money

Checks

Electronic Payment

E-Money

Measuring Money

- How do we measure money? Which particular assets can be called “money”?
- Construct **monetary aggregates** using the concept of liquidity:

Measuring Money

- **Monetary Aggregates**

M1 = Currency
+ Traveler's checks
+ Demand deposits
+ Other checkable deposits
Total M1

M2 = M1
+ Small-denomination time deposits
+ Savings deposits and money market deposit accounts
+ Money market mutual fund shares (retail)
Total M2

Measuring Money

- **Monetary Aggregates**

includes only paper money and coins in the hands of the nonbank public and does not include cash held in ATMs or bank vaults

includes only traveler's checks not issued by banks

includes business checking accounts that do not pay interest as well as traveler's checks issued by banks

includes all other checkable deposits, particularly interest-bearing checking accounts held by households

certificates of deposit with a denomination of less than $100,000 that can only be redeemed at a fixed maturity date without a penalty

non-transactions deposits that can be added or taken out at any time

similar to money market mutual funds, but are issued by banks

Shares are retail accounts on which households can write checks.

- Traveler's checks
  - Check issued by a financial institution which functions as cash but is protected against loss or theft. Traveler's checks are useful when traveling, especially in case of overseas travel when not all credit and debit cards carried by a person will be accepted. A charge or commission is usually incurred when a person exchanges cash for traveler's checks.

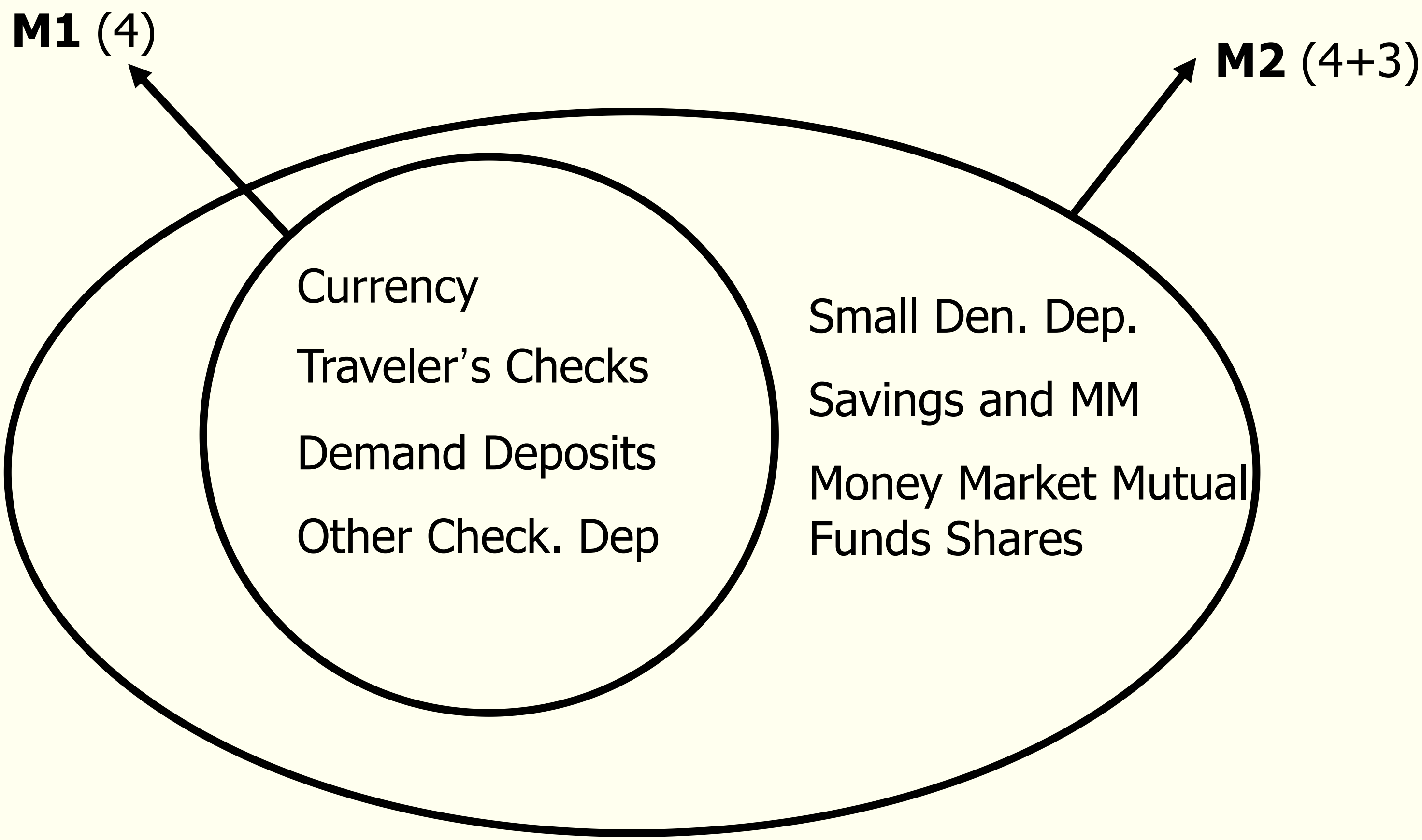
The Federal Reserve's Monetary Aggregates

TABLE 1 Measures of the Monetary Aggregates

	Value as of August 18, 2014, ($ billions)
M1 = Currency	1,206.1
+ Traveler's checks	3.3
+ Demand deposits	1,089.9
+ Other checkable deposits	477.4
Total M1	2,776.7
M2 = M1	
+ Small-denomination time deposits	533.0
+ Savings deposits and money market deposit accounts	7,338.2
+ Money market mutual fund shares (retail)	642.5
Total M2	11,290.4

Source: http://www.federalreserve.gov/releases/h6/hist.

The Federal Reserve's Monetary Aggregates



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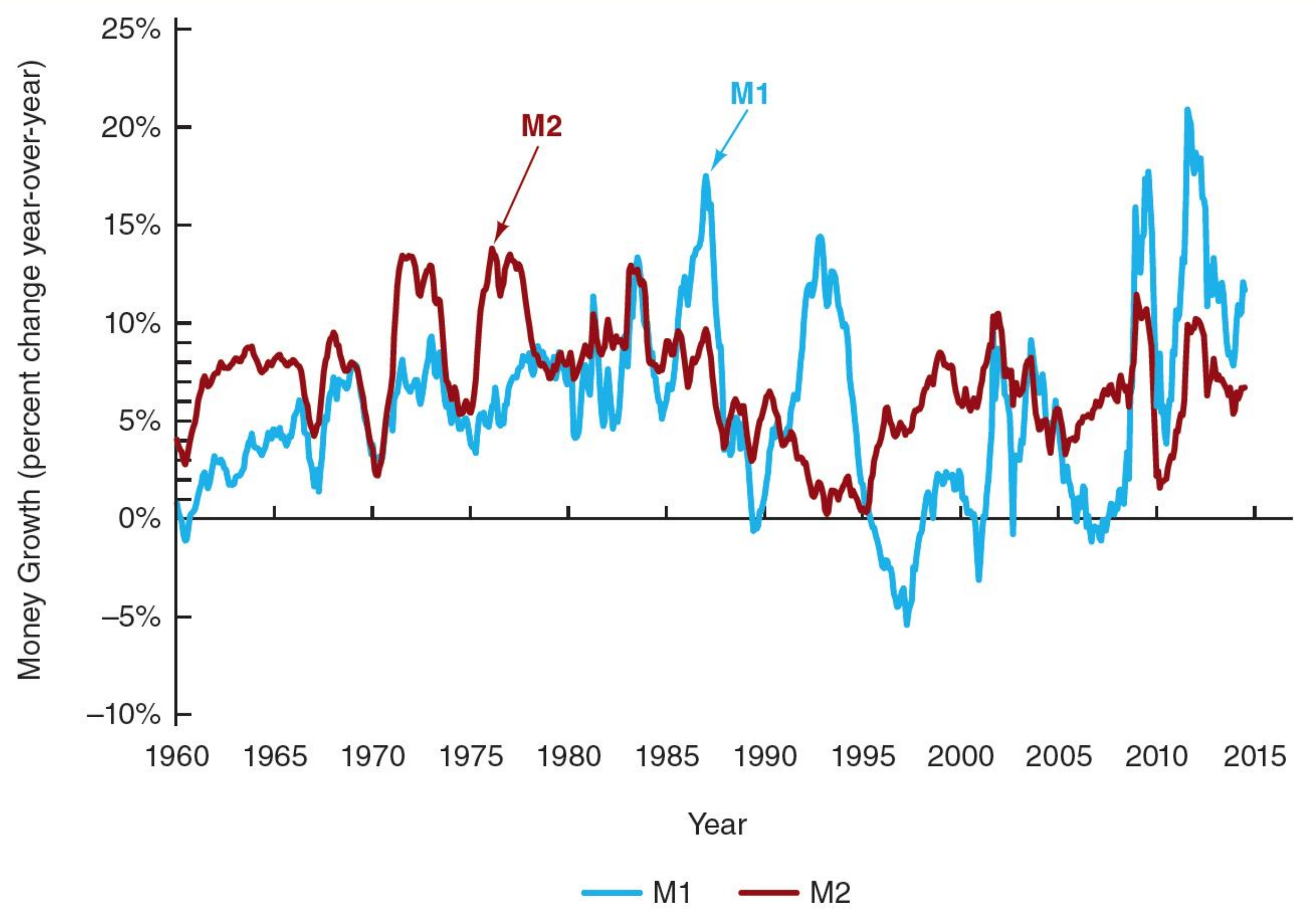
The Federal Reserve's Monetary Aggregates

- M1 versus M2: Does it matter which measure of money is considered?
- M1 and M2 can move in different directions in the short run (see figure).
- Conclusion: the choice of monetary aggregate is important for policymakers.

Measuring Money

- M0：流通中现金；
- M1：M0+企业活期存款；
- M2：M1+准货币（定期存款+居民储蓄存款+其它存款）

Figure 1 Growth Rates of the M1 and M2 Aggregates, 1960–2014



Source: Federal Reserve Bank of St. Louis, FRED database: http://research.stlouisfed.org/fred2

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Exercises

- Currency includes

  A) paper money and coins.

  B) paper money, coins, and checks.

  C) paper money and checks.

  D) paper money, coins, checks, and savings deposits.

Exercises

- To an economist, money is

A) anything that is generally accepted in payment for goods and services or in the repayment of debt.

B) a flow of earnings per unit of time.

C) the total collection of pieces of property that are a store of value.

D) always based on a precious metal like gold or silver.

Exercises

- When we say that money is a stock variable, we mean that

  A) the quantity of money is measured at a given point in time.

  B) we must attach a time period to the measure.

  C) it is sold in the equity market.

  D) money never loses purchasing power.

Exercises

- Ranking assets from most liquid to least liquid, the correct order is

  A) savings; bonds; house; currency.

  B) currency; savings; bonds; house.

  C) currency; house; savings; bonds.

  D) house; savings; bonds; currency.

Exercises

- Paper currency that has been declared legal tender but is not convertible into coins or precious metals is called _________ money.

  A) commodity

  B) fiat

  C) electronic

  D) funny